Joshua Muhumuza

AI Safety Researcher | Ag. Principal Systems Engineer | AI Governance Practitioner

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Profile

AI safety researcher and systems engineer with demonstrated impact across clinical AI deployment, mechanistic interpretability, and AI governance. A Senior systems engineer with over 7 years in designing and developing Enterprise systems that have over 10,000+ users worldwide.

Education

MSc Computer Science (AI Track) — Makerere University *2022 – 2024*

GPA: 4.66/5 - First Class Honors | Sponsored by Google DeepMind

BSc Software Engineering — Makerere University *2014 – 2018*

GPA: 4.78/5 — First Class Honours | Sponsored by Government of Uganda

Professional Certificates

Technical AI Safety — BlueDot Impact *2026*

AI Safety techniques, gaps and implementation to ensure a safety AI ecosystem

Google Professional Cybersecurity — Google *2025*

Cybersecurity risks in systems, exposure and mitigations to solving these problems

Professional Project Management — Google *2025*

Project inititation, planning, running trational and agile projects

Research Projects

AI Safety: Mechanistic Interpretability in Clinical AI *2026 – Present*

- Identified a spurious attention circuit (L8H10) in a Vision Transformer trained on chest X-rays that is causally verified as harmful yet completely invisible to standard output-level safety evaluation, with direct implications for clinical AI procurement and governance in low-resource settings.
- Finding replicates across architectures and datasets. Manuscript submitted to the XAI4Science Workshop @ NeurIPS 2026.

AI Safety: Multilingual LLM Safety Evaluation *2026 – Present*

- Showed that binary refusal-rate metrics used across multilingual jailbreak evaluations are structurally unreliable: across 360 completions from three open LLMs in Swahili, Yoruba, and Luganda, only 2 contained coherent harmful content, with the remaining non-refusals distributed across three distinct benign failure modes conflated by the naive metric.
- Proposed a five-way triage protocol for multilingual safety evaluation. Manuscript submitted to the JUDGe Workshop (Can We Trust the Judge?) @ NeurIPS 2026.

Pluralistic Alignment: Annotation Bias & Hate Speech *2025 – Present*

- Demonstrated empirically that majority-vote annotation in hate speech datasets suppresses minority value judgments at the hate/offensive boundary, a structural failure with direct implications for multilingual, multicultural populations across East Africa.
- Paper Accepted and published at ICML conference under Pluralistic Alignment Workshop @ ICML 2026.

ITU Project: University of Oxford *February 2025 – May 2025*

- Collaborated on country-focused policy case studies (primarily Kenya), analyzing national responses to emerging AI safety institutes and governance structures. (Funded by Institute of Applied Policy)

AMR Surveillance AI Model *October 2024 – Present*

- Served as Tech Lead in designing and piloting a low-resource recommendation models and companion mobile application for antimicrobial resistance (AMR) surveillance, co-infection detection, and real-time information dissemination to community. (Funded By: GIZ).
- Engineered efficient models optimized for low-compute environments to enable accurate monitoring and identification of AMR threats in data.

Sickle Cell Early Detection AI *June 2022 – Present*

- Led development of a lightweight deep learning model for enhanced, accurate interpretation of rapid diagnostic tests in low-resource contexts, improving early sickle cell disease detection. (Funded By: National Institutes of Health grant).
- Optimized computer vision algorithms to boost diagnostic reliability on mobile devices, addressing accessibility challenges in underserved regions.

Tuberculosis Classifier and report Generation AI *Jan 2023 – Sept 2023*

- Directed creation of an explainable multi-task deep learning architecture for chest X-ray classification and automated report generation, overcoming long-context limitations in large language models for clinical usability. (Funded by Lacuna Fund)
- Built an interpretable AI system that streamlines TB screening workflows, with potential to accelerate diagnosis in high-burden settings.

Grants & Funding

RUFORUM Grant: Integrated Learning Management System *$160,000 | 2026*
Regional Universities Forum for Capacity Building in Agriculture (RUFORUM)
- Awarded to design and deploy an integrated learning management system serving 175+ universities across Africa, strengthening shared academic infrastructure at continental scale.

Buganda Royal Institute Grant: Vocational Academic Management System *$20,000 | 2026*
Buganda Royal Institute *Uganda*
- Awarded to develop an academic management system for vocational training institutions, supporting skills development and economic participation across the region.

NIH Grant: Sickle Cell Early Detection *2023 – Present*
National Institutes of Health *USA*
- Co-investigator on NIH-funded project developing lightweight sickle cell detection models for mobile deployment in low-resource clinical settings in Uganda.

AMR Surveillance Grant: ADAPT One Health *2024 – Present*
BFTR and GIZ Germany
- Funded development and deployment of an AMR Mobile Application and dashboard to support the National AMR board in tracking AMR and promote education awareness in Uganda.

Weather & Climate Grant: WIMEA *2024 – Present*
NORHED
- Funded development and deployment of design and monitoring of automatic weather stations in East Africa.

Lacuna Fund Grant: TB Chest X-Ray Classification *2022 – Present*
Lacuna Fund *Remote*
- Funded development and deployment of multi-task deep learning model for TB classification and report generation at Uganda's national referral hospitals.

Work Experience

Ag. Principal Systems Engineer *Jul 2026 – Present*
Makerere University *Kampala, Uganda*
- Manage and oversee the university's core data systems, ensuring data integrity, availability, and continuity across institutional platforms.
- Support the University Management and Appointments Board with data and reports to inform Human Resource decision-making, including staff appointments and promotions.

Senior Systems Engineer *May 2025 – Jun 2026*
Makerere University *Kampala, Uganda*
- Maintained and upgraded existing enterprise systems (MakData, MakARS, RIMS, Mak e-Vote, e-HRMS), ensuring continued reliability for 175+ affiliated institutions.
- Designed and led development of the e-Procurement System, digitising university procurement workflows end-to-end.
- Designed the University Council Business Tracker System, enabling systematic tracking of council resolutions and business items.

Systems Engineer *Jan 2021 – Apr 2025*
Makerere University *Kampala, Uganda*
- Built Makerere University Data Warehouse (MakData), first scalable integration of siloed institutional systems, processing 25TB+ of data and serving 175+ universities across Africa.
- Developed the Academic Records System (MakARS), digitising records from the 1900s and managing 15TB+ of institutional data under the DARP project.
- Developed Graduate e-Supervision System (RIMS), contributing to a 60% increase in on-time graduations; adopted in CAES with pilot expansions to CHUSS and CEES.
- Designed and deployed Mak e-Vote, the university's secure electronic voting platform; managed five consecutive guild elections as appointed oversight committee member.
- Developed Electronic Human Resource Management System (e-HRMS), automating the full HR lifecycle across the institution.

Google DeepMind Scholar *Jan 2022 – Sept 2024*
Google DeepMind *London, UK*
- Designed and fine-tuned ML models for health diagnostics in Uganda; two projects secured NIH and Lacuna Fund grants and were deployed at national referral hospitals.

Lead AI Developer *Dec 2018 - Dec 2020*
Emaisha Ltd *Kampala, Uganda*
- Led team that designed and launched Uganda's first farmer call center, securing the company's inaugural UNCDF grant and expanding agricultural support services.
- Developed Livestock Manager web and Android portal for efficient livestock management.
- Engineered a USSD-based farm management application for low-connectivity areas, promoting digital inclusion and service delivery to underserved farmers.

Software Engineer *Aug 2018 - Dec 2018*
Thinvoid Ltd *Kampala, Uganda*
- Designed and prototyped data management system for NGO Educate!, boosting productivity across 700+ schools in Africa.

Software Developer *Jun 2017 - Jun 2018*
WIMEA-ICT *Makerere University, Uganda*
- Developed web portal for managing automatic weather stations; design adopted by Uganda National Meteorological Association.

Publications & Presentations

Muhumuza, J. et al. (2026). **Spurious Attention Heads in Vision Transformers: Mechanistic Identification of Annotation-Driven Bias in Medical Chest X-Ray Analysis.** *Submitted, XAI4Science Workshop @ NeurIPS 2026.*

Muhumuza, J. et al. (2026). **Refusal Rate Is Not Safety: Three Failure Modes of Multilingual LLM Safety Evaluation.** *Submitted, JUDGe Workshop (Can We Trust the Judge?) @ NeurIPS 2026.*

Muhumuza, J. et al. (2026). **When Majority Vote Silences Minority Values: Pluralistic Alignment Failures in Hate Speech Detection.** *Accepted, Pluralistic Alignment Workshop @ ICML 2026.*

Nsabagwa M., Muhumuza J. et al. (2018). **Minimal Idle Listening and Centralized Scheduling in TSCH Wireless Sensor Networks.** *IEEE, Greece.* doi

Nsabagwa M., Muhumuza J. et al. (2018). **Condition Monitoring and Reporting Framework for WSN-based Automatic Weather Stations.** *IEEE, Botswana.* doi

AI Governance & Policy

Researcher: Kenya AI Safety Institute Framework *Feb – May 2025*

ITU / Oxford University (Institute for AI Policy and Strategy) *Remote*

- Researched the draft policy framework for Kenya's AI Safety Institute, one of the first such institutional frameworks produced for an African context, informing national-level AI governance design.
- Conducted cross-country analysis of AI safety governance structures across EAC partner states, documenting the near-total absence of safety evaluation infrastructure and the institutional risks this poses for deployed AI systems.

AI Strategy Contributor & Digital Leaders Forum Member *2026 – Present*

East African Community / EASTECO *East Africa*

- Active contributor to the EAC's regional AI strategy; participated in the 4th STI AI Conference on AI4EAC alongside policymakers, regulators, and civil society leaders from all EAC partner states, with GIZ as a key international partner.
- Advising on AI safety, equitable deployment, and governance frameworks for a region where AI adoption is accelerating without corresponding safety infrastructure or regulatory capacity.